
COIT20258 Assessment 2

Due date: Week 9 Friday (18 September 2026), 11:59 PM AEST

Weighting: 35%

Length: N/A

1. Objectives

The purpose of this assessment is to assess your skills attributable to the following learning outcomes and achieving the expected graduate attributes of *advanced level knowledge, communication, cognitive, technical, and creative skills, and self-management.*

- Design system architecture and components using design patterns and object-oriented design
- Develop three-tiered software applications using agile and plan-driven methods
- Plan and conduct test-driven development

2. Assessment Task

Your task in this assessment is to review and enhance the requirements specification developed in your assessment one and build the software prototype.

Case Study (from Assignment one)

You are to develop a set of user stories that capture the user requirements for a Disaster Response System (DRS). Disaster response consists of warning/evacuation, search and rescue, providing immediate assistance, assessing damage, continuing assistance, and the immediate restoration or construction of infrastructure, etc.

Different activities may be involved for different disasters. For example, the activities for a typical hurricane response would include assessment, restoration, and demolition of buildings; removal of debris and waste; repairs to land-based and maritime infrastructure; and providing public health services.

This project is to develop a Disaster Response System (DRS) that needs to:

- Allow users to report different kinds of disasters, which could be a hurricane, a fire, an earthquake, etc.
- Provide quick and accurate assessment of the reported disasters and provide prioritized responses accordingly.
- Coordinate different departments and people for efficient & effective responses.

The system would need interfaces to other departments & organizations such as fire and emergency, hospital, electricity, transportation, waste management, water supply, schools, and law enforcement.

You will be developing the software prototype DRS in three stages.

You have completed the *first stage (Assessment One individual)* in which you have completed requirements engineering identifying functional and non-functional requirements through capturing use cases, drawing use case diagrams, and conceptualizing system architecture.

You are now in the *second stage (Assessment Two individual) of DRS*, and you will work individually and build an initial software prototype called DRS-Initial which will be an MVC driven JavaFX Scene builder-based GUI desktop application. You are required to revise your requirements engineering specification incorporating necessary corrections. You will follow the revised requirements engineering specification and develop the required designs and test plans, and implement the software prototype.

While you will be working with the Assessment Two for DRS-Initial, you are required to *include and implement and test two creative features or functionalities* that were not explicitly mentioned in the Assessment Two specifications.

- Identify the classes. You can use your use case diagrams, use cases, and functional requirements to correctly identify the classes. Develop the correct UML class diagrams with attributes, methods, relationships, and cardinalities.
- Correlate the context diagram, use case diagrams, and the attributes of classes to identify the user input and the data to be saved. Create test plan for the classes in UML diagrams towards test driven development using JUnit testing.
- Build the software prototype:
 - designing and revising the enhanced GUI using Scene Builder
 - following Model View Controller pattern
 - implementing the model class and methods
 - implementing the controller classes and methods for events handling

3. Coding

Include necessary accessors, mutators methods, constructors, and toString() method for each class. Also, follow good coding practices, using meaningful names, camel case notation for naming, constants as necessary, and include meaningful comments. You can use NetBeans or another IDE to develop your application. Follow the coding standards given in the Unit website. Once you have created your UML class diagrams with correct attributes, you can use your IDE to generate most of the methods such as constructors, getters, and setters. The auto-generated toString() method will not give correct display format. Write your own toString() method.

4. Report

You should submit a report containing the following details.

1. Detailed UML class diagrams with attributes, methods, data types, and associations.
2. Test plan showing test data and expected results.
3. Your reflection of the overall requirements specification, user interface design, and implementation.

5. Submission

You should submit the following on Moodle via the Assignment Two submission link:

- your report in word document (.doc or .docx).
- a zip file of the NetBeans project (or a zip file of the other IDE you use) with your software prototype.

6. Marking Criteria

No.	Criteria (Total: 35)	Marks Allocated	Marks Awarded
	Coding		
1	Implementation of allowing users to report different kinds of disasters	4	
2	Implementation of providing quick and accurate assessment of the reported disasters and providing prioritized responses accordingly	4	
3	Implementation of coordinating different departments and people for efficient & effective responses	4	
4	Implementation of two additional Creative features/functionalities	4	
	Report		
5	Detailed UML class diagrams with attributes, methods, data types, and associations	4	
6	Test plan showing test data and expected results	5	
7	Your reflection of the overall requirements specification, user interface design, and implementation.	5	
8	Well-presented report with student details, good document structure with sections, paragraphs, tables/figures, and correct spelling, grammar.	5	
	Penalties		
9	Late Penalty (5% of total allocated marks per calendar day)		
10	Plagiarism (penalty as per the plagiarism policy)		
	Total	35	